Aditya Guhagarkar

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Profile: Wireless PHY/DSP engineer focused on end-to-end link-level simulation, RF/channel modeling, modem algorithms, fixed-point DSP, and AI/ML-assisted wireless system analysis.

Education

University of Michigan Aug. 2025 – Dec. 2026 (Expected)
Master of Science, Electrical and Computer Engineering | GPA: 4.00/4.00 (Weighted: 4.13/4.30) *Ann Arbor, MI*

Coursework: Digital Communications and Codes (A+), VLSI for ML and Communications (A), Reinforcement Learning Theory (A+), Digital Communication Theory (A+), Probability and Random Processes (A).

Indian Institute of Technology, Indore Nov. 2021 – May 2025
Bachelor of Technology, Electrical Engineering | GPA: 9.05/10 *Indore, India*

Coursework: Signals and Systems, Probability and Random Processes, Digital Signal Processing, Digital Communications, Information Coding Theory, Communication Systems, IoT Communication Systems (LTE, WiFi, 5G, CDMA, TCP/IP).

Honors: Institute Gold Medal – Best All-Round Graduate (2025)

Technical Skills

Programming & Simulation: Python, C++, MATLAB/Simulink, Simscape, ns-3, GNU Radio
Wireless & Networking: 5G NR PHY, OFDM, MIMO, 3GPP Layer-1 procedures, TCP/IP fundamentals, Wireshark
Hardware & Tools: SDR (ADALM-PLUTO, USRP B210), Vivado, Proteus, PLECS, Arduino, CC3220SF LaunchPad
Certifications: Qualcomm 5G Introductory-Level Certification; Mastering 5G PHY.

Industry Experience

Impinj, Inc. Jun. 2026 – Aug. 2026
Incoming DSP/RF Modeling Intern | Communication Systems Engineering *Seattle, WA*

- Selected to develop RF/DSP simulation infrastructure for RAIN RFID systems on the Communication Systems Engineering team, contributing to a Python-based link-level simulator for research and product development.
- Scoped to design a unified geometry-based RF propagation framework covering free-space loss, reflection, multipath, antenna directionality, and frequency-dependent gain.
- Will improve the fidelity of physical-layer modelling for evaluating next-generation RF/DSP algorithms under realistic propagation, antenna, and multipath conditions.
- Will build unit and integration tests against analytical channel models and document simulation best practices.

Project Experience

MFCC Audio Feature Extraction Accelerator | *ECE 598: VLSI For ML and Comms, Final Project* Fall 2025

- Designed a fixed-point DSP accelerator for audio feature extraction, implementing pre-emphasis, framing/windowing, 512-point FFT, Mel filterbanks, log scaling, DCT, and liftering in Verilog.
- Validated the hardware model against a MATLAB floating-point reference, achieving $< 1\%$ numerical error and $\sim 7.7\ \mu s$ latency per frame for real-time low-power audio processing.
- Gained hands-on experience translating signal-processing algorithms into fixed-point hardware blocks with latency, numerical accuracy, and implementation constraints.

Non-Terrestrial Assisted Relay Selection and Scheduling for Vehicular Networks | *IEEE ANTS 2025* 2025

- Developed a hybrid non-terrestrial vehicular network integrating mmWave V2V, UAV-assisted relaying, and LEO satellite fallback links to maintain connectivity under severe blockage and coverage gaps.
- Formulated joint relay selection and transmission scheduling as a constrained optimization problem to minimize total time slots while meeting throughput thresholds; implemented and benchmarked CP, DQN, and PPO-based schedulers.
- Achieved 206.8 Gbps avg throughput (PPO/CP) with 43–463 slots vs. JRDS 190.9 Gbps, 43–494 slots and TDMA 10 Gbps, 500–2500 slots; DQN reached 201 Gbps with 43–465 slots.

RL for SWIPT and RIS-Aided Scheduling | *Under Dr. Sumit Gautam* Sept. 2024 – Mar. 2025

- Evaluated scheduling strategies for RIS-assisted SWIPT systems with nonlinear energy harvesting, comparing TDMA, random selection, constraint programming (CP), and DQN-based approaches.
- Developed a DQN scheduler achieving $\sim$90% of CP throughput with over 1000$\times$ lower decision latency.

Receiver-Aware Link Design for NB-IoT Uplink over LEO Non-Terrestrial Networks Feb. 2026 – Present

- Built a 3GPP-aligned NB-IoT uplink link-level simulator for LEO NTN capturing Doppler/CFO drift and Rician fading.
- Proposed PT-CRC+, a phase-tracked coherent repetition combining receiver; Monte Carlo results show elimination of BLER error floors seen in naive coherent/non-coherent combining under CFO drift.

Research Experience

6G Flagship | University of Oulu — Apr. 2025 – Present
Research Intern under Prof. Nandana Rajatheva — *Oulu, Finland*

- Proposed ALORA-GNN, an edge-aware feed-forward GNN for QoS-aware downlink power allocation in Massive MIMO under spatially correlated Rician fading with imperfect CSI using deterministic-equivalent signal/interference couplings.
- Enforced exact sum-power feasibility via softmax projection and incorporated per-user QoS guarantees through an unsupervised augmented-Lagrangian objective, eliminating iterative optimization during deployment.
- At $R_{\min} = 40$ Mbps, achieved the highest user-level QoS satisfaction (0.785) with competitive weighted-rate performance and 4.5×–122× lower inference latency than unfolded WMMSE-Net, WMMSE, and dual-QoS WMMSE baselines.

University of British Columbia — May 2023 – Jul. 2023
Research Intern under Prof. David Michelson — *Vancouver, Canada*

- Built an SDR-based channel sounding and Doppler emulation setup using ADALM-PLUTO radios to measure channel impulse response, path loss, Doppler shifts, and link degradation under RF impairments.
- Implemented LTE-based baseband processing and real-time Doppler correction in MATLAB, reducing BER by nearly 100% under controlled test conditions through feedback-based digital compensation.
- Performed measurement-driven debugging of wireless links, connecting propagation effects, RF distortion, and PHY-layer signal processing performance.

University of Michigan — Aug. 2025 – Present
Research Assistant under Prof. Lei Ying — *Ann Arbor, MI*

- Developed a streaming RL framework for online control in a digital-twin-like simulation (non-stationary dynamics, partial observability), demonstrating improved stability and average return vs. multiple DQN baselines.
- Implemented contrastive RL to improve sampling-based planning (RRT), enabling robust trajectory generation for high-DOF systems, relevant to low-latency closed-loop decision-making in networked autonomy.

Teaching Experience

EECS 301: Probabilistic Methods for Engineering, UMich | *Grader, Fall'25, Winter'26* — Aug. 2025 - May. 2026

- Graded weekly probability and stochastic-process assignments for 100+ students across Fall 2025 and Winter 2026.

EECS 455: Wireless Communication Systems, UMich | *Grader, Winter'26* — Jan. 2026 - May. 2026

- Graded weekly assignments for 20+ students in wireless communication systems.

Selected Publications

1. **A. Guhagarkar**, T. Sivalingam, N. Rajapaksha, et al "QoS-Aware Learning-Based Power Allocation for Massive MIMO Under Correlated Rician Fading," *VTC Fall, Recent Results*, 2026. **[Submitted]**
2. **A. Guhagarkar**, T. Sivalingam, V. Bhatia, N. Rajatheva, and M. Latva-aho, "RL-Based Optimization of Relay Selection and Transmission Scheduling for UAV-Aided mmWave Vehicular Networks," in *Proc. WPMC*, 2024. **[Paper]**
3. **A. Guhagarkar**, V. Bhatia, and S. Gautam, "Towards Efficient Scheduling in RIS-Aided Wireless Networks with Non-Linear Energy Harvesting," in *Proc. ICCCNT*, 2025. **[Paper]**
4. A. Navarkar, R. Prajapati, **A. Guhagarkar**, and V. Bhatia, "Efficient Relay Selection and Transmission Scheduling in Non-Terrestrial Assisted Vehicular Networks," in *Proc. IEEE ANTS*, 2025 (Delhi). **[Paper]**

Awards & Honors

MITACS Globalink Fellow (2023) – Awarded a fully funded research internship at the University of British Columbia.

IEEE PES India Scholarship (2022–24) – One of 3 students selected nationally for academic excellence and leadership.

Chess – Internationally rated chess player (FIDE 2046); 2-time SGFI U-14 National Champion; State Champion U-9.